

FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING (25CS1201E)

2025-26, Term-III

Project Problem Statement Submission Form

Sec No-5	Team No-	Project Title- Digital Report Card Generator
Team Lead Roll Number		2520090093
Team Lead Name		Yuktha Alumuri
Team Member Roll Number		2520090002
Team Member Name		Marri Hruthika

1. Abstract

The Digital Report Card Generator is a web-based solution designed to modernize the traditional report card system in the education sector. Schools often face issues like manual errors, time-consuming preparation, poor record management, and limited accessibility for parents, while physical report cards can be lost and are not eco-friendly. This project addresses these challenges by providing a centralized platform where teachers can easily enter subject-wise grades, add comments, and include attendance details through a user-friendly interface.

The system automatically generates well-structured digital report cards, allows PDF downloads, and sends notifications to parents for quick access and review. It also includes features like e-signature for acknowledgment, report history tracking, archival storage, and an admin panel for management. Built using modern technologies such as React.js, Node.js, and REST APIs, the solution improves efficiency, accuracy, and communication while promoting a secure and paperless academic environment.

2. Module Mapping

(Provide details of how modules are divided among team members.)

Module Name	Assigned Roll Number(s)	Signature
Foundations of Front-End Engineering & Framework Design	2520090093	
JavaScript/TypeScript Engineering and React Component Model	2520090002	

3. Module Details

(Explain the responsibility of each student concerning their assigned module.)

2520090093:

Responsible for implementing Foundations of **Front-End Engineering & Framework Design**, including component-driven UI design, Virtual DOM concepts, rendering flow, reusable frontend architecture, responsive layouts, and declarative UI development using React.js. Also worked on framework-based frontend structuring and user interface composition.

2520090002:

Responsible for implementing **JavaScript/TypeScript Engineering and React Component Model**, including ES6+ concepts, state management, props handling, React hooks, component reusability, asynchronous operations, form validations, and frontend integration using React.js and JavaScript. Also handled styling approaches and interactive UI behavior.

4. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

- HTML5
- CSS3
- JavaScript (ES6+)
- React.js
- Node.js
- npm
- VS Code
- Git & GitHub

Guide Name and Signature (remarks if any) _____